

# A Deterministic Brane-Lattice Substrate: Ontological Foundation and Wave Architecture (Paper I)

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## Abstract

We present the ontological foundation and shared mathematical structure for a series of papers investigating a minimal deterministic substrate hypothesis. The substrate is a 4D cubic lattice embedded with codimension zero in a 4D Euclidean ambient; the timelike direction is selected by the Lorentzian sign structure of the brane action, not by the ambient geometry. Its essential nonlinearity is purely geometric: the induced metric contains the coupling  $\partial_i u \partial_j u$  when the embedding is excited transversely, producing lateral contraction without any auxiliary field. We derive the continuum hyperelastic action from the discrete lattice energy, establish the equations of motion, and characterize the linearized wave-branch structure, including the transverse and longitudinal speeds  $c_T^2 = (1 - \alpha)k_s a^2 / m$  and  $c_L^2 = k_s a^2 / m$ . The transverse–lateral coupling is the shared geometric origin of Lorentz, gauge, color, gravity, matter, and mass bridges developed in companion papers; the present paper exports the interface equations each companion imports. We also commit to a retrocausal worldtube interpretation forced by Bell’s theorem under the assumptions of determinism, measurement independence, and a finite-speed substrate: entanglement is the continuity of a branching 4D worldtube, and the arrow of causality is a property of soliton chirality.

**Keywords:** deterministic substrate; brane lattice; geometric nonlinearity; induced metric; hyperelastic action; linearized waves; Bell’s theorem; retrocausal worldtube

## 1. Introduction

The Standard Model and general relativity are extraordinarily successful at organizing experimental facts. At the same time, their combined fundamental inventory—multiple fields, symmetry sectors, symmetry breaking structure, and many parameters fixed by measurement—invites a complementary line of inquiry: whether a smaller set of ground ingredients could underlie this effective complexity. This paper is the first in a series exploring such a *minimal-ingredient reconstruction* program.

**Substrate-first strategy.** We adopt a deterministic *substrate-first* strategy: posit an ontically real wave medium and treat familiar field theories as coarse-grained descriptions of constrained wave dynamics. The goal is *not* to dispute established phenomenology, but to ask what could make gauge structure, relativistic kinematics, and particle-like quantization appear naturally as emergent regularities.

**Ground ingredients (minimal commitments).** The theoretical framework developed here keeps its primitive commitments deliberately small: (i) a single substrate ontologically realised as a 4D cubic lattice embedded with codimension zero in a 4D Euclidean ambient, with the timelike direction picked out by the Lorentzian sign structure of the brane action

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(Section 3.1), and macroscopically described by its continuum long-wavelength theory on a spacelike slice, (ii) pure geometric coupling between displacements perpendicular to a link and the lengthening of that link, and (iii) a controlled separation between a linear wave sector and a nonlinear self-guidance sector. Gauge potentials, relativistic coordinates, and particle labels are then interpreted as emergent bookkeeping attached to stable excitation sectors rather than as independent fundamental substances.

**Geometric source of nonlinearity.** One modeling distinction is crucial throughout this series. The microscopic link law may be linear in the scalar extension, but the extension itself is the full Euclidean distance between neighboring nodes in  $\mathbb{R}^4$ . Equivalently, in the continuum description the induced metric already contains the quadratic term  $\partial_i u \partial_j u$  when the fourth embedding component is excited. The primary nonlinearity of the theory therefore comes from geometry—the Pythagorean dependence of link lengths and the induced metric—not from an added interaction field, an imposed threshold, or a hand-crafted saturation rule. This point matters because the same geometric mechanism is intended to underwrite contraction, self-guidance, and localized particle structure.

**Substrate hypothesis.** We assume a brane lattice with the 4D-in-4D ontology of Section 3.1: a 4D cubic lattice in a symmetric 4D Euclidean ambient, with the timelike direction selected by the brane action’s Lorentzian sign structure. For technical analysis on a spacelike slice the brane is parameterised by an embedding map

$$\mathbf{X} : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^4, \quad (x, t) \mapsto \mathbf{X}(x, t), \quad \Omega \subset \mathbb{R}^3, \quad (1)$$

with  $t$  the slicing parameter along the inside observer’s timelike direction. The dynamics is specified by a local hyperelastic action on  $\Omega$ , constructed from invariants of the induced metric of the embedding. This is the minimal coordinate-free way to write an elastic energy without hard-coding preferred directions into the continuum description; microscopically, a cubic lattice substrate is allowed, and large-scale isotropy is treated as an empirical constraint on admissible lattice couplings and parameter regimes [1–3].

**Transverse–lateral coupling as the shared geometric origin.** A central structural result of this paper is that a single geometric mechanism — the induced-metric term  $\partial_i u \partial_j u$  — underlies all emergence bridges in the series: it sources the contraction channel interpreted as gravity (Section 3.9), sets the amplitude-dependent self-guidance that localizes solitonic modes, and controls the speed splitting between transverse and longitudinal branches (Equation (17)) from which all bridge papers inherit their linearized interface.

**Relation to existing ideas.** The emergence of an effective light cone from a substrate whose time direction is selected by the action signature has close analogues in analogue-gravity systems [4–6]. The present use of the term “brane” is literal (an elastic medium), and should not be confused with field-theoretic brane-world scenarios in high-energy theory [7]. The closest contemporary programme is the quaternionic quantum mechanics of Danielewski, Sapa, and Roth, in which the vacuum is an ideal elastic solid (a Planck–Kleinert crystal), elementary particles are soliton-like standing waves, gravity appears as a density-dependent change in wave speed, and baryons are modelled as three-quark standing-wave complexes [8,9]. We share the elastic-medium ontology and the particles-as-solitons commitment, but differ in three structural choices: the carrier here is a *real* vector substrate on a 4D codimension-zero brane whose time direction is selected by the action signature rather than an external time; the complex and gauge structure is *derived* per wavepacket from band isolation and Berry/Wilczek–Zee holonomy rather than postulated through a quaternion field; and we make no claim to fix fundamental constants from the model.

**Paper genre and contribution.** This paper is the *shared foundation* for the series: it establishes the substrate model, derives the linearized wave-branch exports that each companion paper imports, and commits to the retrocausal worldtube interpretation forced by Bell's theorem. It does not develop any specific bridge in quantitative detail; those treatments appear in the companion papers (Section 2).

**Paper organization.** Section 2 states the series architecture: what this paper exports, what each companion paper imports, and what is not claimed anywhere in the series. Section 3 formulates the substrate model, making the geometric source of the nonlinearity explicit. Section 4 collects the linearized wave-branch structure and states it as a set of interface equations. Section 5 states the Bell boundary condition: a retrocausal worldtube reading in which causality emerges from soliton chirality and entanglement is the continuity of a branching 4D worldtube.

## 2. Series scope and interface architecture

### 2.1. Structure of the series

The emergence program is organized as a series of five papers sharing a common substrate model but developed around logically independent bridges:

1. **Core (this paper).** Ontological foundation, continuum model, linearized wave structure, and the Bell interpretive commitment (Sections 3–5).
2. **Lorentz and gravity bridge.** Imports: branch speeds (17), transverse–lateral coupling (30), dual-observer setup. Derives: the observer-universality argument for effective Lorentz symmetry at speed  $c_T$ ; residual birefringence as a falsifiable signature; quantitative Newtonian reduction of the contraction channel sourced by  $\partial_i u \partial_j u$ .
3. **Gauge and color bridge.** Imports: linearized branch speeds, polarization eigenbundles from the multi-branch wave system, the U(3) triplet structure of the three spacelike components. Derives: Berry connection  $a_\mu = i\langle u | \partial_\mu u \rangle$  as gauge potential  $A_\mu$ ; Berry curvature as field tensor  $F_{\mu\nu}$ ; Wilczek–Zee connection for the degenerate triplet as  $u(3)$ -valued potential; traceless  $\mathfrak{su}(3)$  sector and kinematic confinement of color-neutral combinations.
4. **Matter and mass bridge.** Imports: equations of motion (26), geometric nonlinearity, gauge connection from Paper III. Derives: soliton eigenproblem as a self-adjoint BVP with nonlinear self-guidance; topological identification of stable modes; emergent rest mass as the energy of a self-confined closed-loop wave pattern, without invoking the Higgs mechanism.
5. **The Bell boundary condition (Paper V).** Imports: substrate determinism, locality, and time-symmetry of the brane action (Section 5). Develops: the uniqueness argument that retrocausality is the only completion consistent with this substrate's ontology; causality from soliton chirality; matter and antimatter as forward- and backward-propagating worldtubes; entanglement as a V-branching 4D world-volume; implications for the simulation program.

### 2.2. Interfaces exported by this paper

The following results, derived in Sections 3 and 4, are treated as fixed interfaces by companion papers. Companion papers may cite these labels directly without re-derivation.

- **Discrete lattice action:** Equation (7) (from Section 3.3).
- **Induced metric:** Equation (10) and its near-identity expansion Equation (29).
- **Continuum action and constitutive law:** Equation (24) and Section 3.5.
- **Equations of motion:** Equation (26).
- **Linearized branch speeds:**  $c_L^2 = k_s a^2 / m$  and  $c_T^2 = (1 - \alpha) k_s a^2 / m$  from Equation (17).

- **Long-wavelength dispersion:** Equation (32). 130
- **Regime of validity:** Section 4.2. 131
- **Bell interpretive commitment:** the retrocausal worldtube stance (Section 5), imported by all companion papers without re-derivation; full development in Paper V. 132 133

### 2.3. Non-claims of the series 134

To maintain honesty across the series, the following items are not claimed in any of the five papers as currently scoped: 135 136

- a derivation of the full Standard Model gauge group, generations, or precise coupling constants; 137 138
- a completed quantitative reduction of the contraction channel to a Newtonian limit with a derived value of  $G$ ; 139 140
- an identification of any specific topological object with a specific physical particle — which objects the substrate dynamics stabilizes is an open numerical question; 141 142
- a derivation of the QCD-vs-EM scale hierarchy from the prestress parameter  $\alpha$ :  $\alpha$  sets a coupling ratio in the linear spectrum, but whether a true dynamical scale hierarchy emerges from the nonlinear sector is unsettled. 143 144 145

### 2.4. Bell as a fixed core commitment 146

Bell's theorem is stated in Section 5 as part of this core paper because it constrains the ontological interpretation of the substrate before any specific emergent phenomenon is considered. The retrocausal worldtube reading — in which entanglement is the continuity of a branching 4D worldtube and causality is a property of soliton chirality — is imported by all companion papers as a fixed stance, not a working hypothesis. The full uniqueness argument and the worldtube interpretation are developed in Paper V. 147 148 149 150 151 152

## 3. Substrate model: lattice microphysics and continuum (IR) effective description 153 154

### 3.1. Ontology and kinematics 155

The fundamental object of this theory is a *4D brane lattice embedded in a 4D Euclidean ambient with codimension zero*: the lattice is the substrate, and there is no extra perpendicular direction in the ambient. The ambient  $\mathbb{R}^4$  is fully symmetric — no preferred direction — and serves only as a stage on which Euclidean distances are measured. All asymmetries that we will encounter (the distinction between time and the three spatial directions, the split between gauge and gravity sectors) are properties of the brane and of the inside observer's frame, not of the ambient. 156 157 158 159 160 161 162

Let  $\Omega \subset \mathbb{R}^4$  be the intrinsic (material) brane domain (either a periodic cell  $\Omega = \mathbb{T}^4$  or a bounded hypercube with clamped boundary). The fundamental field is the embedding map 163 164 165

$$\mathbf{X}(x) \in \mathbb{R}^4, \quad x = (x^1, x^2, x^3, x^4) \in \Omega, \quad (2) \quad 166$$

whose components  $X^A(x)$  ( $A = 1, \dots, 4$ ) give the brane configuration in the ambient. The four ambient directions are geometrically equivalent. The asymmetry that picks one of them as the *timelike* direction is a property of the brane action  $S[\mathbf{X}]$ , not of the ambient: the action has Lorentzian sign structure — the standard  $T - V$  Lagrangian, recast on the lattice as link-energy terms with opposite sign for the timelike-direction contributions. Globally, cosmological boundary conditions (a Big Bang vertex emanating along one specific ambient direction) fix this choice unambiguously, so every inside observer agrees on which axis is time. 167 168 169 170 171 172 173 174

**Working spacelike-slice description.** For technical work that does not require explicit 4D treatment — the linear-regime dispersion of Section 4, the gauge-sector diagnostics developed in the gauge bridge companion paper, and the localized-mode analysis of the matter bridge companion paper — it is convenient to slice the brane perpendicular to the timelike direction. Each spacelike slice is parameterised by  $x = (x^1, x^2, x^3) \in \Omega_3 \subset \mathbb{R}^3$ , with the timelike coordinate  $t := x^4$  acting as the slicing parameter. In this description the embedding takes the equivalent form

$$\mathbf{X}(x, t) \in \mathbb{R}^4, \quad x = (x^1, x^2, x^3) \in \Omega_3, \quad t \in \mathbb{R}, \quad (3)$$

that of a three-dimensional material domain embedded in the 4D ambient. The fourth ambient component, denoted  $u$  in this slicing, is the inside observer's chosen timelike component of  $\mathbf{X}$ ; relative to the spacelike slice it plays the role of a transverse “amplitude” direction. With this identification, all the constitutive, kinematic, and gauge-sector developments below carry through verbatim, now understood as describing the inside observer's spacelike slice.

### 3.2. Discrete lattice substrate (microphysical postulate)

The continuum model is treated as the long-wavelength effective theory of a 4D cubic lattice substrate. Lattice sites are labeled by  $n \in \mathbb{Z}^4$  with spacing  $a$  and reference positions  $x_n = a n$ ; each site carries an embedded configuration  $\mathbf{X}_n \in \mathbb{R}^4$ . The discrete brane action is a sum of link-energy contributions with the Lorentzian sign structure of Section 3.1: contributions involving the timelike direction enter with opposite sign from those involving the three spacelike directions.

On a fixed spacelike slice (perpendicular to the timelike direction), sites are labeled by  $n \in \mathbb{Z}^3$  with embedded configurations  $\mathbf{X}_n(t) \in \mathbb{R}^4$  where  $t$  is the slicing parameter. For a spacelike neighbour stencil  $\mathcal{N} \subset \mathbb{Z}^3$ , the slice-restricted potential energy reads

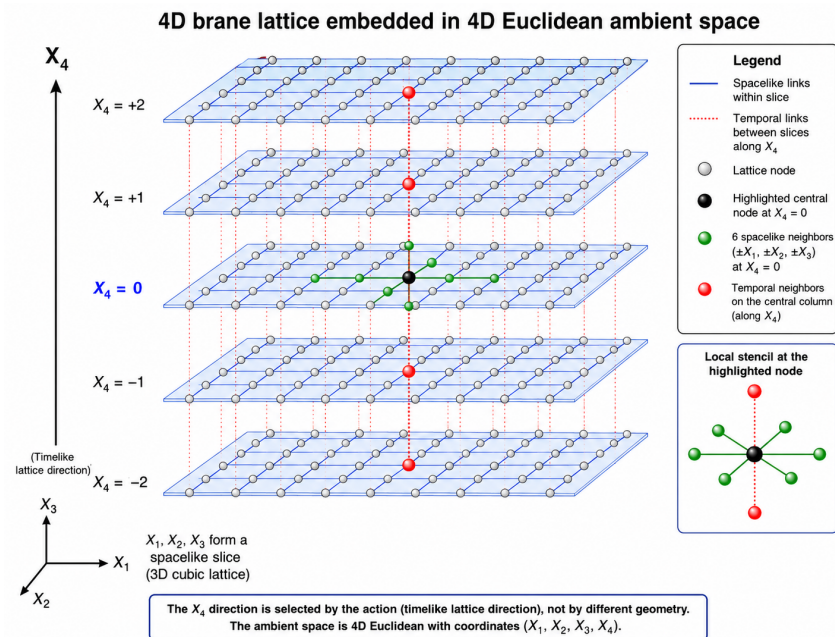
$$U_{\text{lat}}[\{\mathbf{X}_n\}] = \frac{1}{2} \sum_n \sum_{\delta \in \mathcal{N}} k_\delta \left( |\mathbf{X}_{n+\delta} - \mathbf{X}_n| - \alpha a |\delta| \right)^2, \quad (4)$$

with stiffnesses  $k_\delta$  grouped, for example, into axis-aligned, face-diagonal, and body-diagonal shells. The uniform prestretch parameter  $\alpha$  fixes the ratio between the held lattice spacing and the stress-free link length. Link contributions along the timelike direction enter the full brane action with opposite sign and are absorbed into the kinetic term  $\frac{1}{2} \rho_m |\partial_t \mathbf{X}|^2$  of the slice Lagrangian (Equation (23) below); their explicit lattice form is developed in Section 3.3.

**Geometric nonlinearity already present at the microscopic level.** Equation (4) makes an important modeling point explicit. Even if the link energy is Hooke-linear in the scalar extension, its dependence on the node coordinates is nonlinear because the extension is the full Euclidean distance in  $\mathbb{R}^4$ . This Pythagorean nonlinearity is the microscopic origin of the amplitude–lateral coupling that later appears in the continuum as the induced-metric term  $\partial_i u \partial_j u$ . No additional nonlinear interaction field is required for this effect.

**Minimal stencil and lab-frame anisotropy.** The canonical realisation of the lattice in this paper is the 6-neighbor axial-only stencil: each site connects to its six nearest axial neighbours  $\pm \hat{e}_i$ . Diagonal-shell (face- and body-diagonal) bonds are intentionally absent. On this stencil the long-wavelength dispersion is cubically anisotropic at leading order in  $|k|a$ : for example, the longitudinal speed along [100] exceeds the longitudinal speed along [111] by a factor  $1/\sqrt{(3-2\alpha)/3}$ , evaluating to roughly 7.5% at the default operating point  $\alpha = 0.2$ . This anisotropy is not retuned away; it is the structural feature that the





**Figure 1.** The 4D brane lattice embedded in 4D Euclidean ambient space. Each spacelike slice (horizontal planes, labeled by  $X_4$ ) is a 3D cubic lattice with axial-only spacelike links (blue). Red links connect adjacent slices along the timelike direction  $X_4$ ; the timelike lattice direction is selected by the action's Lorentzian sign structure, not by the ambient geometry. Inset: the 7-site local stencil at the highlighted node, consisting of its six spacelike axial neighbors and two temporal neighbors.

gauge-sector emergence story developed in the gauge bridge companion paper ultimately rests on.

**Dual-observer framework: lab anisotropy vs internal isotropy.** The lab (outside) observer knows the lattice and resolves direction-dependent wave speeds. The internal (soliton) observer is built from substrate excitations and, by hypothesis, has rods, clocks, and signal cones that all renormalize with the same direction-dependent local wave speed. Their measured speed of light, defined operationally as the ratio of an emergent rod length to an emergent clock period, is therefore conjectured to be direction-independent in their own units. This is a strong assumption: it promotes “operational isotropy” from a passive consistency condition to the load-bearing mechanism by which Lorentz-like kinematics emerge for internal observers despite manifest lab-frame anisotropy. The decisive empirical test is whether two solitons in different orientations report the same effective metric; this is the central numerical objective of the Lorentz bridge companion paper and is operationalised in the validation roadmap.

**Brillouin zone and geometric-phase diagnostics.** The lattice spacing  $a$  defines a Brillouin zone  $k_i \in [-\pi/a, \pi/a]$ . Because Berry curvature is naturally defined over parameter spaces such as  $\mathbf{k}$ -space eigenbundles, the lattice formulation provides a direct route to Berry-curvature diagnostics on a discretized momentum lattice (see the gauge bridge companion paper).

### 3.3. Explicit discrete 4D brane action

The spacelike-slice description of Section 3.2 treats the timelike direction implicitly, absorbing it into the kinetic term  $\frac{1}{2}m|\dot{R}|^2$ . For the foundational 4D-in-4D ontology of Section 3.1 it is essential to make the timelike direction an explicit lattice link, placing all four spatial directions on equal footing as far as the link topology is concerned and making the Lorentzian sign structure a single explicit choice. Label each node by a spacelike triple

$p = (i, j, k)$  and a timelike index  $l \in \mathbb{Z}$ , with 4D embedded position  $R_p^l \in \mathbb{R}^4$ . The six spacelike axial neighbours form the stencil  $\mathcal{N}_s = \{\pm\hat{e}_x, \pm\hat{e}_y, \pm\hat{e}_z\}$  and the two temporal neighbours form  $\mathcal{N}_t = \{l \pm 1\}$ .

**The action.** The slice-restricted potential is the central-force sum over  $\mathcal{N}_s$ :

$$V^l = \frac{k_s}{4} \sum_p \sum_{\delta \in \mathcal{N}_s} \left( |R_{p+\delta}^l - R_p^l| - \alpha a \right)^2, \quad (5)$$

and the temporal kinetic term on the link  $l \rightarrow l + 1$  is

$$T^{l+\frac{1}{2}} = \sum_p \frac{m}{2} \left( \frac{R_p^{l+1} - R_p^l}{\Delta t} \right)^2. \quad (6)$$

With the Lorentzian sign assignment — spacelike links enter with  $-$  (potential), temporal links enter with  $+$  (kinetic); this is Assumption A5 of the derivation and is *posited*, not derived: it is the property of the brane action that selects one of the four equivalent ambient directions as timelike (backbone #21) — the explicit discrete action is

$$S[R] = \sum_l \Delta t \left( T^{l+\frac{1}{2}} - V^l \right). \quad (7)$$

This is a sum over  $6 + 2$  links per node with a single sign flip for the temporal family.

**Euler–Lagrange identity: Verlet is the variational integrator.** Varying Equation (7) with respect to an interior node  $R_p^l$  (the node appears in  $T^{l\pm\frac{1}{2}}$  and  $V^l$ ) yields

$$m \frac{R_p^{l+1} - 2R_p^l + R_p^{l-1}}{\Delta t^2} = F_p^l := -\frac{\partial V^l}{\partial R_p^l} = k_s \sum_{\delta \in \mathcal{N}_s} \left( |R_{p+\delta}^l - R_p^l| - \alpha a \right) \widehat{(R_{p+\delta}^l - R_p^l)}, \quad (8)$$

which is precisely the Störmer–Verlet update  $R_p^{l+1} = 2R_p^l - R_p^{l-1} + (\Delta t^2/m) F_p^l$ . This is the discrete-mechanics result (Marsden–West): the Euler–Lagrange equations of the discrete action  $S$  are exactly Störmer–Verlet, and the flow is symplectic. Therefore *Verlet is the discrete variational integrator of the action (7)*. The forward initial-value problem (prescribe  $R^0, R^1$ , march  $l \rightarrow l + 1$ ) and the foundational 4D block boundary-value problem (prescribe spacelike-slice data at  $l = 0$  and  $l = N$ , root-find the interior) share *identical local stencil (8)* and differ only in their global boundary conditions.

**Long-wavelength cone speeds.** Taking the continuum limit with  $\rho_m = m/a^3$  and wave vector  $\mathbf{k} = k\hat{e}_x$ , the Christoffel matrix is diagonal on the axial stencil  $\mathcal{N}_s$  (no off-diagonal coupling at any  $\alpha$ ), giving the lattice light-cone speeds

$$c_L^2 = \frac{k_s a^2}{m}, \quad c_T^2 = (1 - \alpha) \frac{k_s a^2}{m}. \quad (9)$$

In dimensionless units  $k_s = a = m = 1$  these reduce to  $c_L = 1$ ,  $c_T = \sqrt{1 - \alpha}$ . At the default operating point  $\alpha = 0.2$  one obtains  $c_T/c_L = \sqrt{0.8} \approx 0.894$ , a residual  $\sim 10.6\%$  longitudinal–transverse gap that is the leading-order consequence of prestretch; it is not retuned away (backbone #8, #15).

**Outlook: foundational block solver.** The Lorentzian action (7) is a saddle (unbounded below), so a foundational block solver must root-find  $\nabla S = 0$  rather than minimize  $S$ . The timelike link is a central-force spring like the three spacelike links, so the substrate is one 4D-isotropic spring lattice parameterized by the temporal rest length  $r_t$ :  $r_t = 0$  is the

linear/Verlet limit used throughout this paper, and  $r_t = \alpha \beta \Delta t$  is the prestressed substrate carrying a timelike geometric quartic. The well-posedness of the two-time boundary-value problem (resolved in the linear regime: a two-past-slice chiral boundary condition has an  $N$ -independent, bounded condition number) and the consistency of a single 4D prestress  $\alpha_t = \alpha$  are tracked outside the manuscript.

### 3.4. Induced metric and isotropic reference metric

The embedding induces a (time-dependent) Riemannian metric on the brane:

$$g_{ij}(x, t) := \partial_i \mathbf{X}(x, t) \cdot \partial_j \mathbf{X}(x, t), \quad i, j = 1, 2, 3. \quad (10)$$

This definition is invariant under rigid motions  $\mathbf{X} \mapsto Q\mathbf{X} + a$  with  $Q \in O(4)$  and  $a \in \mathbb{R}^4$ . Rotational invariance within the brane is obtained by constructing the stored energy density from invariants of  $g$  relative to an isotropic reference metric.

We choose a homogeneous, isotropic reference metric

$$g_{ij}^0 = \ell_0^2 \delta_{ij}, \quad (11)$$

where  $\ell_0 > 0$  is the continuum analogue of a discrete spring rest length. Define the mixed relative metric

$$\hat{g} := (g^0)^{-1} g. \quad (12)$$

Any isotropic hyperelastic energy density can be written as  $W = W(\hat{g})$ , equivalently as a function of the principal stretches or invariants of  $\hat{g}$ .

**Continuum analogue of pre-tension.** The reference metric  $g^0$  encodes the stress-free geometric scale. If boundary conditions enforce a held ground-state scale  $h_*$  (uniform), then

$$\alpha := \frac{\ell_0}{h_*} \in (0, 1] \quad (13)$$

measures the mismatch between stress-free scale and held scale. For  $\alpha < 1$  the brane is uniformly pre-stressed (continuum “pre-tension”). This mismatch controls how strongly transverse gradients backreact on in-brane stresses through the geometric nonlinearity of  $g_{ij}(\mathbf{X})$ . In the lattice substrate picture (Section 3.2), the same  $\alpha$  multiplies the geometric neighbor distances to set spring rest lengths; Equation (13) is the continuum encoding of that uniform prestretch.

### 3.5. Constitutive law: central-force spring and its continuum expansion

**The exact constitutive law.** The substrate’s stored energy is determined entirely by the central-force spring potential. On the 6-neighbor axial-only stencil  $\mathcal{N}_6 = \{\pm \hat{e}_i\}$ , the exact slice-restricted potential (Equation (5)) is

$$V = \frac{k_s}{4} \sum_p \sum_{\delta \in \mathcal{N}_6} \left( |R_{p+\delta} - R_p| - \alpha a \right)^2. \quad (14)$$

This is the energy the simulation kernel computes exactly. It is *frame-indifferent*: it depends only on the Euclidean link length  $|R_{p+\delta} - R_p|$ , not on rigid motions of the ambient. The induced-metric line element  $|\Delta R|^2 = |\partial_i \mathbf{X} dx^i|^2$  is the discrete counterpart of the Riemannian pull-back (10): both measure the physical distance between neighboring material points, confirming that (14) is the natural elastic energy for an embedded lattice.



**Exact link decomposition.** Writing  $E_{\text{link}} = \frac{1}{2}k_s(|\Delta R| - \alpha a)^2$  and splitting into the squared-norm and the Euclidean-norm terms,

$$E_{\text{link}} = \underbrace{\frac{1}{2}k_s|\Delta R|^2}_{\text{exactly quadratic}} - \underbrace{k_s\alpha a|\Delta R|}_{\text{entire anharmonic sector}} + \frac{1}{2}k_s\alpha^2 a^2. \quad (15)$$

The squared-norm term  $\frac{1}{2}k_s|\Delta R|^2$  is a degree-2 polynomial in node coordinates and contributes *only* to the quadratic (linear-elasticity) sector. Every cubic and higher term in the energy descends from the Euclidean-norm term  $-k_s\alpha a|\Delta R|$ , which carries an explicit factor  $\alpha$ . Consequently, *all anharmonic physics vanishes as  $\alpha \rightarrow 0$*  (backbone #17).

**Continuum expansion:**  $W = W_2 + W_4 + \dots$  In the long-wavelength regime ( $a|k| \ll 1$ ,  $|\nabla u| \ll 1$ ) the exact spring energy coarse-grains to a gradient expansion  $W = W_2 + W_4 + \dots$ .

**Leading term  $W_2$  (linear elasticity).** Using the quadratic link strain  $(s_\delta^2)^{(2)} = \alpha(\hat{\delta} \cdot \Delta u)^2 + (1 - \alpha)|\Delta u|^2$  and the axial tensor sums  $T_{ab}^{(2)} = 2\delta_{ab}$ ,  $T_{abcd}^{(4)} = 2Q_{abcd}$  (where  $Q_{abcd} = 1$  iff  $a = b = c = d$ ), the acoustic tensor on  $\mathcal{N}_6$  evaluates to

$$A_{abcd} = \frac{k_s}{a} [\alpha Q_{abcd} + (1 - \alpha) \delta_{ac} \delta_{bd}]. \quad (16)$$

The Christoffel matrix along  $k = k\hat{e}_1$  is diagonal, giving

$$c_L^2 = \frac{k_s a^2}{m}, \quad c_T^2 = (1 - \alpha) \frac{k_s a^2}{m}, \quad (17)$$

in exact agreement with the lattice closed-form dispersion (cross-checked at every  $\alpha$ ). The effective Lamé parameters on the 6-neighbor stencil are

$$\mu(\alpha) = \frac{(1 - \alpha) k_s}{a}, \quad \lambda = C_{1122} = 0. \quad (18)$$

The vanishing of  $\lambda$  and the off-diagonal entries of  $A_{abcd}$  confirm that the 6-neighbor stencil is structurally *cubically anisotropic* ( $C_{1111} - C_{1122} - 2C_{1212} = (2\alpha - 1)k_s/a \neq 0$  for  $\alpha \neq \frac{1}{2}$ ): this is a feature, not a bug — it is the structural source of the gauge sector (backbone #8, #15, #16, #19).

**Cubic term  $W_3 = 0$ .** On purely longitudinal modes ( $\Delta u_\perp = 0$ ) the link length is  $|\Delta R| = |a + \Delta u_\parallel|$ , which is exactly Hookean — no odd anharmonic terms. On purely transverse modes the displacement  $q \equiv |\Delta u_\perp|$  appears only through  $q^2$ , so  $W_3 = 0$  by parity. *The geometric nonlinearity is intrinsically transverse.*

**Quartic term  $W_4$  (soliton stabilizer).** Expanding the Euclidean-norm term on the pure-transverse slice ( $\Delta u_\parallel = 0$ ,  $q = |\Delta u_\perp|$ ):

$$|\Delta R| = \sqrt{a^2 + q^2} = a + \frac{q^2}{2a} - \frac{q^4}{8a^3} + \dots, \quad (19)$$

and collecting the quartic contribution from  $-k_s\alpha a|\Delta R| \supset + (k_s\alpha/8a^2)q^4$ , the continuum coarse-graining gives

$$W_4 = \frac{k_s\alpha}{8a} |\partial u_\perp|^4, \quad (20)$$

with coefficient  $\propto \alpha$  that *vanishes at  $\alpha \rightarrow 0$*  and is positive (geometric hardening,  $Q > 0$ , backbone #17). In 4D the transverse component includes both the lateral and the timelike displacement channels, producing the lateral-amplitude geometric quartic

$(k_s \alpha / 4a) |\partial u^{\text{lat}}|^2 (\partial u^{X^4})^2$  that stabilizes solitons against Derrick (uniform-rescaling) collapse (backbone #17). This is the standard Skyrme balance:  $W_2 \sim \lambda^{-1}$  and  $W_4 \sim \lambda^{+1}$  under  $u(x) \mapsto u(\lambda x)$  in 3D — here  $\lambda$  denotes the Derrick rescaling factor, not the Lamé parameter (which vanishes on this stencil) — with an independent UV cutoff at the lattice spacing  $a$ .

**Green–Lagrange strain and the StVK quadratic-order proxy.** For analytic comparison with the textbook literature it is useful to introduce the Green–Lagrange strain tensor

$$E := \frac{1}{2}(\hat{g} - I) \quad (\text{with } I \text{ the } 3 \times 3 \text{ identity}), \quad (21)$$

and the St. Venant–Kirchhoff (StVK) stored energy density

$$W_{\text{StVK}}(E) = \mu \operatorname{tr}(E^2) + \frac{\lambda}{2} (\operatorname{tr} E)^2. \quad (22)$$

*StVK is a quadratic-order proxy for the central-force spring, not the substrate's constitutive law.* At quadratic order in  $\nabla u$ , StVK reproduces the same wave speeds (17) and the same  $\mu, \lambda$  as the spring (18): the two descriptions are indistinguishable in the linear-elasticity regime of Section 4. At quartic order, however, they diverge critically: the spring's geometric quartic (20) has coefficient  $\propto k_s \alpha / a$ , whereas a naive StVK identification locks the quartic to  $\mu \propto (1 - \alpha)$  and *inverts* the  $\alpha$ -scaling. *Quantitative soliton sizing (matter bridge companion paper) must use the spring law (14)–(20), not StVK.*

On the 6-neighbor stencil the vanishing of  $\lambda$  (Equation (18)) gives Poisson ratio  $\nu = 0$  structurally.

**Compression non-coercivity.** The central-force spring (14) is not polyconvex nor coercive in compression: as a link collapses ( $|R_{p+\delta} - R_p| \rightarrow 0$ ), the pair energy  $\frac{1}{2} k_s (\alpha a)^2$  remains bounded rather than diverging. The lattice spacing  $a$  masks this in practice (configurations cannot compress below the grid), but the continuum limit exposes it. For the linear and moderate-strain results of Section 4 this is immaterial; for the strongly localized modes of the matter bridge companion paper it must be controlled — either by verifying that the relevant configurations remain non-inverting, or by substituting a polyconvex hyperelastic energy (e.g. neo-Hookean, Mooney–Rivlin, Ogden-type) within the same induced-metric framework, which leaves the geometric mechanism of transverse–lateral backreaction unchanged. The geometric quartic (20) that stabilizes solitons against Derrick collapse is a distinct safeguard and does not by itself remove this local compression instability.

**Spacelike-slice versus 4D placement.** The strain (21) is a spacelike-slice object:  $E$  is  $3 \times 3$  and is referred to the Euclidean reference metric of the slice. Its placement in the foundational 4D action (Section 3.3) follows the single timelike-link model: the spring potential  $V$  (14) covers the spacelike links and the timelike link enters with the opposite sign, reducing in the  $r_t = 0$  limit to a separate kinetic term so that  $S = \int (T - V)$  is ordinary elastodynamics. The Lorentzian signature resides in the *sign structure of the action* (Section 3.1), not in the ambient or induced metric: the ambient is Euclidean, so  $g_{\mu\nu} = \partial_\mu \mathbf{X} \cdot \partial_\nu \mathbf{X}$  is positive-definite and the timelike direction is distinguished only by the opposite sign with which its contribution enters  $S$ .

### 3.6. Lagrangian, action, and equations of motion

Let  $\rho_m > 0$  be the brane mass density ( $\text{kg}/\text{m}^3$ ). The Lagrangian density is

$$\mathcal{L}(\mathbf{X}, \partial_t \mathbf{X}, \nabla \mathbf{X}) = \frac{\rho_m}{2} |\partial_t \mathbf{X}|^2 - W(g(\mathbf{X}); g^0). \quad (23)$$

The action is

$$S[\mathbf{X}] = \int_{\mathbb{R}} dt \int_{\Omega} d^3x \mathcal{L}. \quad (24)$$

Define the first Piola stress (as a momentum conjugate to  $\partial_i X^A$ )

$$P_A^i := \frac{\partial W}{\partial(\partial_i X^A)}. \quad (25)$$

For energies depending on  $g_{ij}$  one can write  $P_A^i = 2S^{ij}\partial_j X^A$  with a symmetric second Piola-type tensor  $S^{ij} = \partial W / \partial g_{ij}$ . The Euler–Lagrange equations take divergence form

$$\rho_m \partial_{tt} X^A = \partial_i P_A^i, \quad A = 1, \dots, 4, \quad (26)$$

showing explicitly that isotropy (absence of built-in preferred directions) follows from writing  $W$  in terms of invariants of the induced metric relative to the reference metric  $g^0$ .

### 3.7. Spherical material coordinates (coordinate change only)

**Important: this is only a coordinate change.** All dynamical statements remain those of the coordinate-free formulation  $\mathbf{X} : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^4$  with induced metric  $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$ . We introduce spherical coordinates on the *intrinsic/material* domain as a convenient chart for localized excitations; this does *not* constrain the embedding  $\mathbf{X}$ .

Let  $(r, \theta, \varphi)$  be the standard spherical chart on (a subset of)  $\mathbb{R}^3$ , with Jacobian  $J(r, \theta, \varphi) = r^2 \sin \theta$ . Writing the intrinsic coordinate as  $q^a \in \{r, \theta, \varphi\}$  and denoting derivatives in this chart by  $\partial_a$ , the equations of motion become

$$\rho_m \partial_{tt} X^A = \frac{1}{J(q)} \partial_a \left( J(q) P_A^a \right), \quad A = 1, \dots, 4, \quad (27)$$

which is exactly the Cartesian divergence form (26) written with the standard Jacobian factor.

### 3.8. Linearization and polarized branches

Linearize around a homogeneous background embedding  $\mathbf{X}_*(x) = (x, 0)$  and small displacements  $\mathbf{u}(x, t) = \mathbf{X}(x, t) - \mathbf{X}_*(x)$ . Under standard regularity assumptions on  $W$ , one obtains a linear wave system with (in general) multiple polarization branches, including transverse and longitudinal modes. For later use, we emphasize two generic features: (i) different branches may have different characteristic speeds (and may become dispersive at shorter wavelengths), and (ii) for  $\alpha < 1$  the pre-stress couples transverse excursions (in the embedding) back into in-brane stresses through the induced metric (10). These branches provide the substrate carriers for the geometric-phase and soliton constructions developed below.

### 3.9. Transverse–lateral coupling and an emergent “gravity” channel

A distinctive feature of an embedded elastic medium is that transverse gradients change intrinsic distances. To make this explicit without removing in-brane dynamics, we use a near-identity decomposition around the homogeneous background embedding  $\mathbf{X}_*(x) = (h_* x, 0)$ :

$$\mathbf{X}(x, t) = (h_* x^1 + \xi^1(x, t), h_* x^2 + \xi^2(x, t), h_* x^3 + \xi^3(x, t), u(x, t)), \quad \vec{\xi}(x, t) \in \mathbb{R}^3. \quad (28)$$

The induced metric is then

$$\begin{aligned} g_{ij} &= \partial_i \mathbf{X} \cdot \partial_j \mathbf{X} \\ &= (h_* \delta_i^a + \partial_i \xi^a)(h_* \delta_j^a + \partial_j \xi^a) + \partial_i u \partial_j u \\ &= h_*^2 \delta_{ij} + h_* (\partial_i \xi_j + \partial_j \xi_i) + \partial_i \vec{\xi} \cdot \partial_j \vec{\xi} + \partial_i u \partial_j u. \end{aligned} \quad (29)$$

Setting  $\vec{\xi} \equiv 0$  recovers the familiar graph picture used only for intuition; the “gravity” channel, however, relies on the allowed in-brane relaxation  $\vec{\xi} \neq 0$  sourced by the quadratic term  $\partial_i u \partial_j u$ .

In the small-slope regime, the in-brane strain has the schematic form

$$\varepsilon_{ij} \approx \frac{1}{2} (\partial_i \xi_j + \partial_j \xi_i) + \frac{1}{2} \partial_i u \partial_j u, \quad (30)$$

so the quasi-static balance  $\partial_i \sigma_{ij} \approx 0$  with  $\sigma_{ij} = \lambda \text{tr}(\varepsilon) \delta_{ij} + 2\mu \varepsilon_{ij}$  makes it explicit that the term  $\partial_i u \partial_j u$  acts as a forcing/eigenstrain-like source for the slow in-brane field  $\vec{\xi}$ .

**Pythagorean lengthening and lateral contraction (informal picture).** For a small material displacement  $dx$ , the ambient line element of the graph embedding satisfies

$$ds^2 = h_*^2 |dx|^2 + (\nabla u \cdot dx)^2. \quad (31)$$

Thus any region with nonzero slope  $\nabla u$  necessarily *lengthens* intrinsic distances compared to the flat configuration. In a discrete spring-lattice realization this is the familiar Pythagorean effect  $|\Delta \mathbf{X}| = \sqrt{|\Delta x|^2 + (\Delta u)^2}$ : transverse variation increases spring length even when lateral coordinates are unchanged. In the actual dynamics the in-brane displacement  $\vec{\xi}$  is not constrained; under pre-tension ( $\alpha < 1$ ) the additional metric contribution  $\partial_i u \partial_j u$  acts as an effective source in the in-brane equilibrium equations, producing a compensating contraction/dilation pattern rather than a purely local increase of tensile stress. If the medium is pre-tensioned ( $\alpha < 1$ ), the additional length is paid for by increased tensile stress, which produces net lateral forces that pull material points *toward* the region of transverse excitation. For a localized bump or rotating transverse pattern, the surrounding brane therefore develops an inward bias (contraction) rather than an outward pressure.

In this qualitative sense, the “gravitational mass” of an excitation is tied to the elastic energy stored in transverse gradients (and, more generally, to embedded deformation energy), while the “gravitational field” is the slowly varying in-brane strain/dilation pattern sourced by that energy. The sign is fixed by geometry: increasing transverse arc length increases elastic energy and therefore increases tension, which pulls laterally inward.

With homogeneous pre-stress ( $\alpha < 1$ ), this geometrically induced strain backreacts on the in-brane stress field through the constitutive law, effectively coupling transverse excitation energy into slow modulations of the in-brane configuration.

**Unification across all four directions.** The Pythagorean argument above is direction-blind: in the underlying 4D ontology (Section 3.1) any displacement perpendicular to a link stretches that link, regardless of which ambient direction is perpendicular. The slice-projected description above treats  $u = X^4$  specially because the inside observer has already chosen  $X^4$  as the timelike direction. The same mechanism, applied symmetrically across the four directions, predicts that displacements in the three *spacelike* directions (i.e. in-brane kinetic energy / spatial motion) also lengthen the link along the timelike direction — which is gravitational *time dilation* sourced by stress-energy. Length contraction (timelike-direction excitation stretches spacelike links) and time dilation (spacelike-direction excitation stretches the timelike link) are therefore two faces of one rule. The slice-restricted

analysis of this subsection captures the first face; a full 4D treatment of the brane action would express both within one expression.

This provides a concrete mechanism for a long-range channel often described informally as “gravity” in the present program: localized transverse gradients can act as sources of slowly varying background strain (contraction/dilation), and linear wave packets propagating on top of that background experience branch-dependent effective coefficients (cf. Section 4). In a weak-field, slowly varying regime one may therefore seek a Newtonian-limit reduction in which a scalar potential built from coarse-grained strain (or elastic energy density) obeys a Poisson-type equation. Deriving such a reduction quantitatively (and relating the resulting coupling constant to  $G$ ) is an open problem of the present theory.

#### 4. Wave-branch structure and characteristic speeds

The substrate carries Lorentzian sign structure only at the level of the brane *action* (Section 3.1); the ambient  $\mathbb{R}^4$  is symmetric Euclidean. This section collects the linearized dispersion results derived in Section 3 and states them as exported interface equations for companion papers. The deeper question of whether inside observers built from the same substrate inherit universal effective Lorentz kinematics is the subject of the Lorentz bridge paper and is not claimed here.

##### 4.1. Linearized dispersion and branch speeds

Linearizing Equation (26) about a homogeneous background yields a multi-branch wave system. The characteristic speeds  $c_L^2 = k_s a^2 / m$  and  $c_T^2 = (1 - \alpha) k_s a^2 / m$  are derived in Section 3 from two independent starting points — the closed-form dispersion of the discrete action (Equation (9)) and the acoustic-tensor eigenvalues of the  $W_2$  continuum limit (Equation (17)) — and their agreement there is a self-consistency check across levels of description. For each polarization branch  $b$  one has the long-wavelength dispersion

$$\omega_b^2(\mathbf{k}) \approx c_b^2 |\mathbf{k}|^2 \quad \text{for } |\mathbf{k}| \rightarrow 0, \quad (32)$$

which defines an effective wave cone for that branch.

Two structural points follow immediately. First, the prestress parameter  $\alpha$  controls the *splitting* between transverse and longitudinal speeds:  $\alpha = 0$  collapses them to a single speed  $c_L$ , while  $\alpha \rightarrow 1$  drives  $c_T \rightarrow 0$ , approaching a degenerate shear-free medium. Second, the transverse branch carries the geometric coupling  $\partial_i u \partial_j u$  as its leading nonlinearity (Section 3), which is absent at linear order but feeds back at finite amplitude — the mechanism underlying the gravity-like contraction channel (Section 3.9).

##### 4.2. Regime of validity

The single-branch description (32) is valid when: (i) a single branch dominates transport and mode mixing is negligible, (ii) dispersion at shorter wavelengths is small, and (iii) geometric nonlinearity at larger amplitudes is not yet important. In the present theory, departures from these conditions are not treated as flaws; they are structural consequences of a substrate whose timelike direction is selected by the brane action rather than imposed as a metric primitive, and they delimit the domain where a single-branch or relativistic effective description is expected to apply.

**Interface to the Lorentz bridge paper.** The Lorentz bridge paper imports Equations (17) and (32) and the dual-observer argument (Section 3.9) to ask whether inside observers built from the same substrate inherit universal effective Lorentz kinematics at the wave cone defined by  $c_T$ . Residual birefringence ( $c_L \neq c_T$ ) is then the primary falsifiable signature.



## 5. The Bell interpretive commitment

The substrate of this paper is deterministic (the equations of motion (26) are second-order hyperbolic PDEs) and local (all forces propagate through nearest-neighbor spring contacts at finite speed  $c_T$  or  $c_L$ ). Bell's theorem then implies that the joint distribution of hidden variables and detector settings cannot factorize: measurement independence must be relaxed. Superdeterminism is excluded by the substrate's lack of any mechanism to coordinate microscopic initial conditions with macroscopic measurement choices at space-like separation. The unique remaining option consistent with this ontology is *retrocausality*: future measurement contexts propagate backward along soliton worldtubes to a shared V-vertex in the past, without any spacelike signal.

The brane action is time-symmetric: its Lorentzian sign structure selects a kind of direction (timelike vs spacelike) but not a direction along the timelike axis. The arrow of causality is therefore a property of soliton solutions, not of the substrate. Matter solitons propagate forward; antimatter solitons propagate backward, realizing the Feynman–Stueckelberg interpretation at the soliton level. Entanglement is the continuity of a single branching 4D world-volume that splits at a V-vertex in the shared past of the entangled pair.

This retrocausal worldtube stance is imported by all companion papers as a fixed ontological commitment. A full development of the Bell constraint, the uniqueness argument, and the worldtube interpretation is given in Paper V [10].

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## Appendix A. Symbol dictionary

|                                        |                                                                                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Omega \subset \mathbb{R}^3$          | Intrinsic/material domain of the brane (coordinates $x = (x^1, x^2, x^3)$ ).                                                                                                                           |
| $\mathbf{X}(x, t) \in \mathbb{R}^4$    | Embedding field of the brane into ambient 4D Euclidean space.                                                                                                                                          |
| $t$                                    | Slicing parameter along the inside observer's timelike direction ( $t := x^4$ ); not an external/ambient time (the ambient is symmetric 4D Euclidean, Section 3.1).                                    |
| $g_{ij}$                               | Induced metric: $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$ .                                                                                                                         |
| $g_{ij}^0$                             | Isotropic reference metric: $g_{ij}^0 = \ell_0^2 \delta_{ij}$ (stress-free scale $\ell_0$ ).                                                                                                           |
| $h_\star$                              | Held ground-state scale (e.g. imposed by boundary conditions).                                                                                                                                         |
| $\alpha$                               | Dimensionless pre-stress parameter: $\alpha = \ell_0/h_\star \in (0, 1]$ .                                                                                                                             |
| $\widehat{g}$                          | Mixed relative metric: $\widehat{g} = (g^0)^{-1}g$ .                                                                                                                                                   |
| $E$                                    | Green–Lagrange strain: $E = \frac{1}{2}(\widehat{g} - I)$ .                                                                                                                                            |
| $\lambda, \mu$                         | Effective Lamé parameters derived from the spring constitutive law: $\mu(\alpha) = (1 - \alpha)k_s/a$ , $\lambda = 0$ on the 6-neighbor stencil. StVK agrees at quadratic order only; see Section 3.5. |
| $\rho_m$                               | Brane mass density in the Lagrangian.                                                                                                                                                                  |
| $W(g; g^0)$                            | Isotropic stored-energy density (hyperelastic).                                                                                                                                                        |
| $P_A^i$                                | First Piola stress: $P_A^i = \partial W / \partial (\partial_i X^A)$ .                                                                                                                                 |
| $\mathbf{u}$                           | Displacement field relative to a background: $\mathbf{u} = \mathbf{X} - \mathbf{X}_\star$ .                                                                                                            |
| $u$                                    | Shorthand for a transverse component in a graph-type embedding, $\mathbf{X} \approx (x, u)$ .                                                                                                          |
| $\epsilon$                             | Normalized polarization state of a narrowband multi-component wave.                                                                                                                                    |
| $a_\mu$                                | Berry (rank-1) connection: $a_\mu = i\langle u   \partial_\mu u \rangle$ .                                                                                                                             |
| $f_{\mu\nu}$                           | Berry curvature: $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$ .                                                                                                                              |
| $\mathcal{A}_\mu$                      | Wilczek–Zee connection on an $n$ -dimensional polarization subspace.                                                                                                                                   |
| $\mathcal{F}_{\mu\nu}$                 | Wilczek–Zee curvature: $\partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - i[\mathcal{A}_\mu, \mathcal{A}_\nu]$ .                                                                           |
| $\gamma_\Gamma$                        | Geometric phase / holonomy around loop $\Gamma$ : $\gamma_\Gamma = \oint_\Gamma a_\mu dx^\mu$ .                                                                                                        |
| $\Omega_{\text{mix}}$                  | Mode-mixing (non-adiabatic conversion) rate between nearby polarization bands.                                                                                                                         |
| $\Delta\omega_{\text{gap}}$            | Local spectral gap to the nearest competing polarization band.                                                                                                                                         |
| $a$                                    | Lattice spacing of the cubic substrate (microphysical length scale).                                                                                                                                   |
| $\tilde{a}$                            | Momentum-lattice spacing in the discretized Brillouin zone (typically $\tilde{a} \sim 2\pi/(N_s a)$ ).                                                                                                 |
| $\mathbf{k}$                           | Crystal momentum / Brillouin-zone coordinate for lattice eigenmodes.                                                                                                                                   |
| $\Psi = (\psi_x, \psi_y, \psi_z)^\top$ | Axis-aligned complex narrowband triplet carrier (cubic lattice internal sector).                                                                                                                       |
| $T^a$                                  | Generators of $\mathfrak{su}(3)$ (e.g. Gell–Mann basis) used to decompose a $U(3)$ connection.                                                                                                         |
| $R_p^l \in \mathbb{R}^4$               | 4D node position in the explicit lattice: $p = (i, j, k)$ spacelike triple, $l \in \mathbb{Z}$ timelike index.                                                                                         |
| $k_s$                                  | Central-force spring stiffness of the 6-neighbor spacelike stencil.                                                                                                                                    |
| $m$                                    | Node mass (discrete lattice; continuum mass density $\rho_m = m/a^3$ ).                                                                                                                                |
| $\Delta t$                             | Temporal lattice step (timelike link length in the explicit 4D lattice).                                                                                                                               |
| $\mathcal{N}_s$                        | 6-neighbor axial spacelike stencil: $\{\pm\hat{e}_x, \pm\hat{e}_y, \pm\hat{e}_z\}$ .                                                                                                                   |
| $\mathcal{N}_t$                        | Temporal neighbor stencil: $\{l \pm 1\}$ .                                                                                                                                                             |
| $c_L$                                  | Long-wavelength longitudinal wave speed: $c_L^2 = k_s a^2 / m$ .                                                                                                                                       |
| $c_T$                                  | Long-wavelength transverse wave speed: $c_T^2 = (1 - \alpha) k_s a^2 / m$ .                                                                                                                            |

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